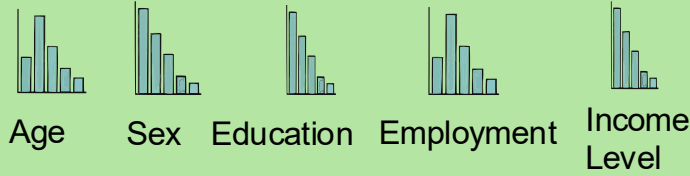
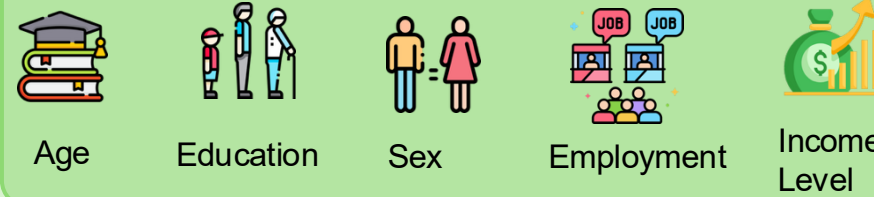


Step1: Regional Targets, Census Summaries and Spatial Data

External Conditions Extracted from Census



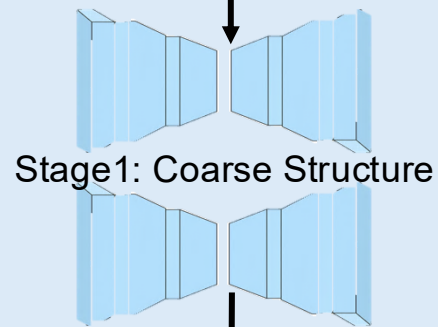
Regional Targets Extracted from PUMS Microdata



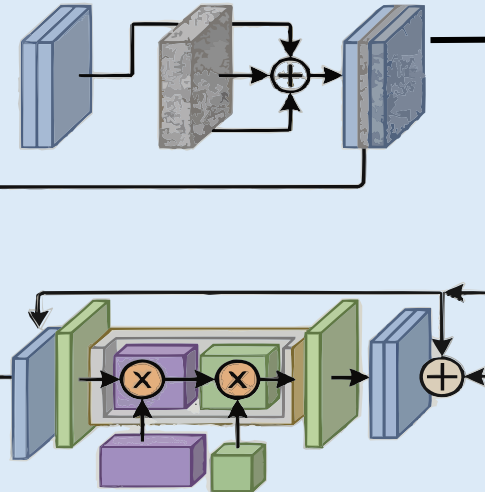
Spatial Support Data



Step2: Hierarchical Joint Distribution Learning



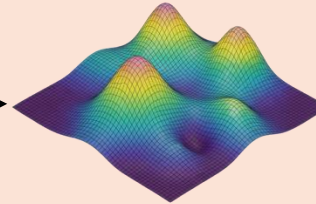
Stage2: Full-joint Diffusion Model



Coarse Regional Distribution

Step3: Synthetic Population Generation and Location Assignment

PUMA-level 5-attribute Full Joint



Sample

Synthetic Population with 5 attributes
1. Age
2. Sex
3. Education
4. Employment
5. Earnings



Point-level Synthetic Population Data

Assign Locations

Spatial Location Model